

Xingjian Wang

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EDUCATION

University of Wisconsin–Madison	BS in Computer Science	Aug 2014 – Aug 2018
Binghamton University	MS in Computer Science	Aug 2018 – May 2020

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java, C++, C

Web Technologies: HTML5, CSS3, React, Next.js, FastAPI, WebSocket, Node.js, Redux, Electron, Tauri, Playwright, Jest, Vite, Web Components, PostgreSQL, Drizzle

Web 3D: three.js, canvas, react-spring, react-three-fiber, @react-three/drei, framer-motion

AI Agent Engineering: Agent orchestration (multi-agent orchestration), LangChain, LangSmith tracing, RAG knowledge base construction, Vector DB retrieval, LLM API integration, agent toolchain design, prompt engineering, context engineering

WORK EXPERIENCE

Beijing Huacheng Technology Co., Ltd. – Zopia AI, Beijing Mar 2026 – Aug 2026

Full-Stack Engineer – AI Short Drama Native Application

Front-end and back-end development, LLM integration, and building the technical foundation for the automated generation of short-drama content.

1. Obra/Superpowers Automated AI Agent Dev Workflow (Claude Code CLI / Copilot CLI)

- Obra/Superpowers is a multi-subagent orchestration AI development framework with native composable Skills: brainstorm for requirement ideation, systematic debugging, code read/write, Git operations, Shell command execution, unit testing, static code review, file diff comparison, dependency analysis. It can execute end-to-end engineering tasks automatically within isolated environments.
- Extended custom Skills on top of the native framework: Figma design parsing, feature solution evaluation (in-house implementation vs third-party API adoption), development cost & timeline estimation, empowering agents to make early-stage technical design decisions.
- Feature development workflow: Input product requirements + technical constraints → Agent orchestrates Skills to decompose tasks automatically, subagents implement frontend components and backend APIs in parallel, completing coding, self-testing and pre-code review within isolated Git workspaces.
- My role: Own requirement validation and high-level architectural decisions, intervene to resolve engineering roadblocks beyond agent capabilities, conduct final code review and mitigate release risks. Avoid heavy manual business coding, accelerate parallel delivery of frontend and backend features, and reduce logical defects in AI-generated code.

2. Image Generation Models & Baidu Cloud Media Capability Integration

Provides intelligent visual content support for the Zopia AI short-drama creation platform, covering video storyboard design, character creation, asset generation, and image enhancement; also provides a standalone video subtitle-removal capability that helps users complete video editing and content creation efficiently.

- **Azure Foundry GPT-Image-1.5 & GPT-Image-2.0 integration:** designed and implemented the AzureFoundryGPTImageExecutor image generation executor supporting both text-to-image and image-to-image modes with multiple resolution and quality levels; implemented a precise token-based billing system for cost control; completed seamless integration with the existing task management system.
- **Baidu Cloud Nano Banana model integration:** selected Baidu Cloud as the proxy access channel for Gemini because call costs are far lower than direct access to the official Google API, while domestic network connectivity is more stable.
 - Designed and implemented the BaiduYunNanoBananaExecutor Gemini proxy image generation executor, compatible with native API protocols through a proxy pattern, supporting both Gemini 3 Pro high-quality generation and Gemini 3.1 Flash high-speed inference.
 - Built a manual multi-vendor switch mechanism across Baidu Cloud/Google/RH, enabling instant manual traffic failover if any vendor's service becomes abnormal, improving fault tolerance.
 - Coordinated with the Baidu Cloud team on API integration and interface latency optimization; identified a 6–7 second high-latency bottleneck on the original link where client requests reached the Baidu Cloud gateway.
- **Baidu Cloud video subtitle-removal integration:** designed and implemented the BaiduYunDesubtitleExecutor for intelligent video subtitle erasure.
 - Implemented the complete task flow: create task → upload video to Baidu Cloud VOD → call the subtitle-removal API → poll the task after submission (queryDesubtitleTask) → transfer the result video to S3 → delete temporary files on Baidu Cloud.
 - Supports global subtitle and specific mode-type configuration, covering various subtitle-removal scenarios.

3. AI Short-Drama Intelligent Agent R&D and Iteration – based on the agent reasoning model gpt5_4 (Azure OpenAI GPT-5.4)

- **Standardized agent architecture:** unified agent development conventions following the “AgentDef model + system instruction + dedicated toolset” architecture, with each agent mapped to a single production stage so responsibilities are cleanly separated with clear logical boundaries, keeping the short-drama production pipeline standardized, traceable, and iterable.
- **LangChain-based multi-agent supervisor collaboration:** designed and implemented SupervisorRouter, an agent routing and distribution module that centrally manages task dispatch across the automated short-drama production workflow. Replacing traditional keyword matching with a dual-dimension decision mechanism based on user intent (User Prompt) plus the real structured state of the workspace (basicSetting), it determines the current task transition node from the actual inventory of screenplays, character assets, and storyboard data in the database, and dispatches each task to the designated SubAgent. It retains the reasoning process and target agent routing results, enabling automatic switching among the screenplay creation, character design, and storyboard rendering agents, with interpretable and traceable execution.
- **Screenplay Agent (screenplay_writer) iteration:** content-first, data-traceable design philosophy; constrains screenplay creation standards and CRUD business strategies through a customized high-precision system prompt (heavy prompt system). Integrated the core toolset of save_user_screenplay, write_file, and edit_file, with screenplay data persistently written via Drizzle DB. Real-time front-end UI refresh is triggered through global IO events; all screenplay modifications go through targeted tool operations, avoiding implicit edits and ensuring deterministic, traceable screenplay data.
 - Built the Agentic RAG sub-agent for the screenplay agent: constructed a short-drama vertical knowledge base on the ChromaDB vector database with the text-embedding-3-small embedding model, and used gpt-4o-mini (temperature = 0.2) for retrieval-augmented generation. Supports semantic retrieval of screenplays and project assets, continuously supplying the main screenplay agent with historical materials and reference context, effectively improving content consistency and asset reuse efficiency.
- **Character Designer Agent (character_designer) iteration:** a pipeline that converts screenplay semantics into visual assets, with a full toolset covering AI image generation, prompt-optimization guidance, and multi-view image generation. The agent connects to the image generation task management system and creates asynchronous image generation tasks through TaskManager. Entity tables serve as the single source of truth for character, scene, and prop resources, unifying visual style at the source and building a solid foundation of visual consistency for downstream storyboarding and video generation.
- **Storyboard Artist Agent (storyboard_artist) iteration and upgrade:** optimized the existing pluggable dynamic routing architecture to give users the ability to choose video generation models and generation methods; refined the multi-mode scheduling logic across Seedance HD video, start_frame/end_frame keyframes, and the Sora full pipeline. Deeply integrated the Baidu Cloud video subtitle-removal post-processing capability, iterated on the end-to-end video generation pipeline, and closed the gap on subtitle artifact repair in AI-generated videos.

4. Product Experience Iteration & Internationalization

- Fixed image rendering anomalies in the chat box, optimized image display interactions, and added click-to-enlarge and hover-zoom preview capabilities to improve the chat experience.
- Iterated on the subscription plan comparison table, adding a unit-price display module with dynamic discount calculation and automatic USD/CNY dual-currency conversion, improving table rendering and pricing transparency.
- Built a global language selector supporting 12 languages and internationalized 216 files, fully covering multilingual usage scenarios.
- Developed a video timeline editing feature supporting video clip deletion and drag-and-drop reordering, improving the visual editing experience.

Beijing Cytoderm Technology Co., Ltd., Beijing

Jul 2025 – Mar 2026

Frontend Engineer

Cybopal One (right robotic arm product) website migration — automated website sync Claude Agent

Company product Cybopal One website architecture upgrade: moved off the frontend built on the Shopify online store, retaining only the Shopify backend APIs and rebuilding with the Next.js App Router. The marketing team delivered an empty-shell React app as the prototype for the website design — with no API integration, not Next.js-compatible, and not ready for production. Manual migration would require batch modification of components, routes, and styles, with heavy, error-prone repetitive work.

Independently designed and developed an AI-driven automated sync Orchestrator built on the Anthropic Claude Agent SDK, achieving fully automated engineering-grade migration from a React app to a Next.js app:

1. Engineering automation capabilities

- Parsed package.json to auto-install missing dependencies and filter redundant toolchain packages; batch-synced assets to the Next.js public directory;
- Auto-mapped page files to Next.js App Router paths; merged global styles, configured root routes, and output a migration summary.

2. Smart File Transformer Agent

- Intelligently detected and injected “use client” directives;
- Automatically replaced react-router-dom with Next.js native routing (Link/useRouter);
- One-click compatible conversion of route attributes and navigation logic; preserved business code integrity with non-invasive modifications.

3. Agent orchestration layer (Orchestrator)

Main control flow orchestration that automates the entire migration pipeline — dependency comparison and installation → static asset sync (images, text, etc.) → AI page conversion → root route configuration → global style merge — with no manual intervention.

Cytoderm (left robotic arm product) website: independently built a complete frontend system from 0 to 1

- Independently built the product website (React + Vite), responsible for the entire process from project initialization to launch, making it the primary external showcase for the left-arm product.
- Built the company's internal Design System (React Storybook) covering core components including Header, Footer, Button, Input, Color System, and Icons; designed unified component APIs, interaction behaviors, and responsive specs, and published it as a private NPM package to provide a consistent visual and interaction foundation across multiple products.
- Integrated cross-framework animation capabilities: selected dynamic effects from Vue Bits (<https://vue-bits.dev/>), re-encapsulated them as Web Components (customElements API), and seamlessly embedded them into the React website for background motion and visual presentation, notably enhancing the website's tech feel and product showcase.

CyboPal-One (right robotic arm) Linux dual-screen interactive application development (react-three-fiber / @react-three/drei / react-spring / Three.js)

1. Companion 3D character & multimodal interaction system Pauli – small screen Demo

- Designed and implemented the companion 3D character Pauli as the core visual and interaction carrier of the small screen on the dual-screen robotic arm, building its character state machine, behavior system, and interaction logic.
- Implemented multi-state character animation and behavior responses based on React Three Fiber, covering auto-tracking, voice, and focus modes; linked voice interaction, gesture recognition, and head tracking with 3D animation to strengthen immediate visual feedback under multimodal input.
- Implemented dynamic character expressions across lifecycle states (idle, interaction, sleep) — blinking, looking around, and falling asleep after long inactivity — enhancing the sense of companionship and liveness.
- Designed and implemented mesh morphing based on vertex deformation, supporting natural transitions between forms (e.g., sphere ↔ bean shape, dynamically growing a hat) with controlled deformation while preserving topology.
- Built a procedural animation system with react-spring for smooth effects such as boot-in and shutdown-out farewells, position switching, and character size transitions.
- Implemented a logging system that traces user input events (WebSocket events for gestures, voice, button clicks), character animation triggers (bounce, flip, morph), and state machine transitions (mode switching, expression changes), with log export on both Electron and browser, facilitating issue localization and behavior reproduction.
- Applied multiple Three.js rendering performance optimizations: useMemo for geometry caching, useRef for animation state to reduce re-renders, frameLockedRef for early exit from the render loop, and direct BufferAttribute vertex data manipulation, ensuring a smooth 60fps experience.

Intuit, Mountain View, USA

Mar 2024 – May 2025

(Global leading financial software company; flagship products: TurboTax, QuickBooks, Mailchimp)

Software Engineer 2 – Front End

React | Intuit AppFabric | Redux | TypeScript | Jest | Enzyme | Webpack | Node.js

Improved the internal developer experience for teams using the IDX (data exchange) feature. Built and refactored the core data exchange widgets — key parts of the banking interaction experience such as bank search (e.g., ProviderSelection) and connection creation/management (e.g., CreateConnection, EditConnection). Developed and maintained the testing and mocking platform that lets product teams efficiently test multiple component flows through a single interface; these platforms simulate the real runtime behavior of components in major applications such as TurboTax, QuickBooks, Mailchimp, and Credit Karma, accelerating other engineers' development processes.

Playground testing platform development and maintenance

1. Intuit AppFabric iFrame integration

Used Intuit's AppFabric iFrame to replace Zoid (PayPal's framework), establishing a standard integration pattern for embedding Intuit content into third-party (non-Intuit) applications.

2. Core component development – BulkReuseConnection & BulkAcquire

A key TurboTax feature that improves user experience by reusing bank account connections from Credit Karma. It allows TurboTax users to reuse existing connections to import files, simplifying the tax-filing process.

3. Debugging and testing in container testing activities

Participated in widget container testing activities, debugging and testing platform functionality. Fixed critical issues such as the close button not displaying correctly and resolved UI alignment problems (e.g., loading animation position offset).

4. Production environment testing enhancement for Playground

Added a production environment testing option to Playground, allowing developers to test/simulate under production conditions without accessing external environments (previously only end-to-end testing environments were supported).

5. Input parameter persistence for Playground

Added functionality so that input parameters in Playground remain unchanged after page refresh or reload, plus the ability to reset parameters back to their default values.

6. Load time measurement and logging

Measured the load times of widget containers and their flow-config and added logging for performance monitoring.

Developer experience renewal program

1. Removed custom CSS overrides

Removed custom CSS overrides targeting IDS (Intuit Design System) components to ensure composable components follow minimal styling principles.

2. Style standardization with IDS (Intuit Design System)

Replaced hard-coded CSS values with design system variables (e.g., `var(--container-border-primary)`) so styles adapt automatically to different themes (e.g., TurboTax, QuickBooks), avoiding manual updates.

3. ContentOverrides content configuration management

Added configuration options to decouple content management (e.g., text, images) from code logic, so team members can update content through remote configuration files without modifying code.

4. Component UI redesign and upgrade

Led the UI redesign of all components, improving user interaction experience and UI consistency across the Intuit product ecosystem. Also introduced process-level feature flags to switch between old and new components, ensuring system stability.

Grubhub: Food Delivery, New York, USA

Aug 2021 – Jan 2024

(One of the largest food delivery platforms in the US)

Software Engineer 1 – Front End

React | Redux | TypeScript | TestCafe | Python | FastAPI | Redis

- In the convenience store development team, led the development of new Post-Purchase Experience features and page enhancements, significantly improving click-through and conversion rates for convenience store shops (7-Eleven, GrubGoods).
- Collaborated with cross-functional teams to design and develop a RESTful API for restaurant menus using Python and FastAPI.
- In the Diner Application team within the Core Diner vertical, primarily built and maintained frontend infrastructure: component migration, building base UI components, and fixing unit test and end-to-end (E2E) test issues.
- Developed and implemented a complete analytics system (Analytics-service) based on Google Analytics and Clickstream, covering user behavior tracking, event monitoring, and data collection across web and mobile platforms, with a focus on e-commerce core metrics such as cart interactions, checkout flows, and user authentication.
- Built and maintained multiple frontend services, including clickstream-service, pageview-service, cart-service, cart-data-service, and giftcard-service.
- Deeply investigated Google Analytics / Clickstream E2E test failures and continuously fixed them until all passed.

Key achievements:

1. Convenience Store UI Enhancement

- Led the design of new restaurant menu templates, category carousels, and section headers for convenience store pages; enabled users to quickly browse a selection of products on the homepage and jump to full category pages by clicking section headers.
- Developed and added a new high-conversion post-purchase experience banner on the Order Confirmation page, raising the click-through rate from 3% to 8% and significantly improving user engagement and conversion.
- Completed a full UI redesign of the convenience store experience with dedicated single-category pages, raising the conversion rate from 1.5% to 4.0%.
- Implemented virtual scrolling (virtual-scroll) for convenience store category pages to efficiently handle large volumes of data or product listings, improving page performance and user experience.

2. Menu Service Redesign

- Played a key role in the menu service redesign, driving a more flexible, dynamic menu structure.
- Adopted a microservices architecture to split the original monolithic menu service into smaller, focused services, each responsible for one aspect of menu management such as product catalog, pricing, and availability.
- Introduced caching to reduce database calls, significantly improving system performance and reducing peak-hour service response times by approximately 40%.

(The largest arts and crafts retail chain in the US)

Software Engineer 1 – Front End

Participated in the web and mobile application development of Michaels' new e-commerce platform. Built the seller platform supporting sellers in building and managing online stores, products, orders, payments, and more. Implemented the user-to-seller registration flow.

- **Frontend work:** built cross-browser pages using React, Redux, and HTML/CSS/JS; developed core seller-side pages and interaction components, combined with Chakra-UI / MUI to improve UI consistency and development efficiency.
- **Backend work:** implemented CRUD APIs for seller data and product systems based on Node.js (Express/Sails) and MongoDB/S3; designed database models and wrote unit and integration tests.